



Algebra & Trigonometry:

An Unabridged Intuitive Exposition and Lecture Notes

Complete 13-Chapter Rigorous Reference Manual

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Algebra & Trigonometry: Complete Engineering Foundations

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Engineered in Japan. Published by the Burmese Artificial Intelligence Research Institute. This unindexed textbook manual provides full structural foundations for computer vision, robotics kinematics, and machine learning matrices. No ISBN has been registered.

Author Profile

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1 | Prerequisites: Foundational Algebra Essentials

To understand complex non-linear mathematical frameworks, we must first master the underlying number systems and operations that govern algebraic expressions.

1.1 Real Numbers: Taxonomy and Structural Axioms

Real numbers ($\mathbb{R}$) form the foundational continuous space upon which computational metrics are mapped.

Hierarchical Breakdown of Number Systems

The real number line is structured as a collection of nested sets:

- **Natural Numbers ($\mathbb{N}$):** $\{1, 2, 3, \dots\}$ — The discrete positive counting integers.
- **Integers ($\mathbb{Z}$):** $\{\dots, -2, -1, 0, 1, 2, \dots\}$ — Introduces negative states and the additive identity (0).
- **Rational Numbers ($\mathbb{Q}$):** $\{\frac{p}{q} \mid p, q \in \mathbb{Z}, q \neq 0\}$ — Numbers that can be expressed as fractional ratios. Their decimal representations either terminate or repeat indefinitely.
- **Irrational Numbers ($\mathbb{I}$):** $\mathbb{R} \setminus \mathbb{Q}$ — Real numbers that cannot be written as simple fractions, characterized by non-terminating, non-repeating decimal expansions (e.g., $\pi \approx 3.14159$, $e \approx 2.71828$, and $\sqrt{2}$).

1.2 Exponents and Scientific Notation

Exponents quantify repeated scaling operations. Let $a, b \in \mathbb{R}$ and $m, n \in \mathbb{Z}$. The structural properties governing exponential mappings are:

$$a^m \cdot a^n = a^{m+n} \quad (\text{Product Rule}) \quad (1.1)$$

$$\frac{a^m}{a^n} = a^{m-n} \quad (a \neq 0) \quad (\text{Quotient Rule}) \quad (1.2)$$

$$(a^m)^n = a^{m \cdot n} \quad (\text{Power Rule}) \quad (1.3)$$

$$(ab)^m = a^m b^m \quad (\text{Distributive Power Property}) \quad (1.4)$$

$$a^{-n} = \frac{1}{a^n} \quad (a \neq 0) \quad (\text{Negative Exponent Inverse}) \quad (1.5)$$

1.3 Radicals and Rational Exponents

Radical operations compute the inverse roots of exponential scaling. For any real value a and positive integer n , the principle n -th root satisfies:

$$\sqrt[n]{a} = a^{\frac{1}{n}} \quad (1.6)$$

When extending this definition to general rational exponents where $m, n \in \mathbb{Z}^+$:

$$a^{\frac{m}{n}} = (\sqrt[n]{a})^m = \sqrt[n]{a^m} \quad (1.7)$$

1.4 Polynomial Frameworks and Factoring

A univariate polynomial is a linear combination of continuous variable components bound by non-negative integer powers:

$$P(x) = a_n x^n + a_{n-1} x^{n-1} + \cdots + a_1 x + a_0 \quad (a_n \neq 0) \quad (1.8)$$

Factoring reverses polynomial expansion by deconstructing expressions into irreducible linear or quadratic terms (e.g., difference of squares: $a^2 - b^2 = (a - b)(a + b)$).

2 | Equations and Inequalities

Solving an equation or inequality means identifying the exact domain limits that satisfy a given mathematical relation.

2.1 Linear Equations in One Variable

A first-degree linear relationship describes flat, unaccelerated spaces. Its standard form is:

$$ax + b = 0 \quad (a \neq 0) \quad (2.1)$$

By isolating the unknown variable, we find its singular unique root: $x = -\frac{b}{a}$.

2.2 Quadratic Equations: Derivation and Complex Solutions

Quadratic equations introduce degree-2 polynomial accelerations, forming parabolic curves.

Definition 1. *The canonical quadratic formulation is expressed as:*

$$ax^2 + bx + c = 0 \quad (a, b, c \in \mathbb{R}, a \neq 0) \quad (2.2)$$

2.2.1 Derivation of the Quadratic Formula

By applying the technique of completing the square to the canonical form, we can isolate x :

$$\begin{aligned} x^2 + \frac{b}{a}x &= -\frac{c}{a} \\ x^2 + \frac{b}{a}x + \left(\frac{b}{2a}\right)^2 &= \left(\frac{b}{2a}\right)^2 - \frac{c}{a} \\ \left(x + \frac{b}{2a}\right)^2 &= \frac{b^2 - 4ac}{4a^2} \\ x + \frac{b}{2a} &= \frac{\pm\sqrt{b^2 - 4ac}}{2a} \implies x = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a} \end{aligned}$$

2.2.2 The Discriminant Mapping

The value inside the square root is called the discriminant ($\Delta = b^2 - 4ac$). It fully determines the geometric properties of the solution space:

- $\Delta > 0$: Two distinct, real roots where the curve crosses the horizontal axis twice.
- $\Delta = 0$: One repeated real root; the parabola's vertex touches the axis perfectly at one point.
- $\Delta < 0$: Two distinct complex conjugate roots ($x = \alpha \pm \beta i$), meaning the graph does not cross the real horizontal axis.

2.3 Linear Inequalities and Interval Notation

Inequalities define bounded solution spaces rather than single coordinate points.

The Inversion Axiom

Multiplying or dividing an inequality by a negative value flips the logical direction of the inequality sign:

$$-kx < \theta \implies x > -\frac{\theta}{k} \quad (k > 0) \quad (2.3)$$

3 | Linear and Quadratic Functions

Functions formalize predictive systems by translating input values into reliable, unique outputs.

3.1 The Functional Dependency Standard

A function f maps elements from an input domain D to an output range R . For each unique input parameter x , it produces exactly one output parameter y . This relationship is written as $y = f(x)$.

3.2 Linear Functions and Slope Dynamics

Linear functions produce constant-rate-of-change straight lines:

$$f(x) = mx + b \quad (3.1)$$

The slope parameter $m = \frac{y_2 - y_1}{x_2 - x_1}$ measures the vertical change relative to the horizontal advance ($\frac{\Delta y}{\Delta x}$), while b defines the y -intercept coordinate $(0, b)$.

3.3 Quadratic Functions: Vertex Optimization

Quadratic functions model non-linear path curves. Writing them in vertex form clearly reveals their key structural features:

$$f(x) = a(x - h)^2 + k \quad (a \neq 0) \quad (3.2)$$

The critical extreme point (h, k) is the vertex of the parabola. Its coordinates can be derived from the standard polynomial form using:

$$h = -\frac{b}{2a}, \quad k = f(h) \quad (3.3)$$

If $a > 0$, the parabola opens upward, making the vertex a global minimum. If $a < 0$, it opens downward, locking the vertex as a global maximum.

4 | Polynomial and Rational Functions

High-degree polynomial and rational functions introduce complex bends, turning points, and asymptotic boundary limits.

4.1 High-Degree Polynomial Systems

For polynomial functions where degree $n \geq 3$, the global behavior at the extreme ends of the graph is dominated entirely by the leading term $a_n x^n$. An n -th degree polynomial features at most n real roots and at most $n - 1$ local extrema turning points.

4.2 Rational Functions and Asymptotic Boundary Limits

Rational functions map the ratio between two distinct polynomial expressions:

$$f(x) = \frac{P(x)}{Q(x)} \quad (Q(x) \neq 0) \quad (4.1)$$

- **Vertical Asymptotes:** Formed at the input values where the reduced denominator drops to $Q(x) = 0$, causing the function's output to explode toward infinity ($\pm\infty$).
- **Horizontal Asymptotes:** Define the stable output level the function approaches as the input variable grows infinitely large ($x \rightarrow \pm\infty$). If $\deg(P) = \deg(Q)$, this boundary stabilizes at the ratio of the leading coefficients: $y = \frac{a_n}{b_m}$.

5 | Exponential and Logarithmic Matrices

Exponential and logarithmic models describe rapid, compounding growth or decay processes.

5.1 Exponential Growth and Decay Models

Exponential models position the independent variable within the exponent power layer:

$$f(x) = a \cdot b^x \quad (b > 0, b \neq 1) \quad (5.1)$$

Continuous growth systems utilize Euler's constant ($e \approx 2.71828$) as their base, expressed as $f(x) = a \cdot e^{kx}$. If $k > 0$, the model tracks exponential acceleration; if $k < 0$, it models exponential decay.

5.2 Logarithmic Conversions and Structural Rules

Logarithms act as the exact mathematical inverse operators of exponential functions, isolating the exponent power layer:

$$y = b^x \iff \log_b(y) = x \quad (5.2)$$

Core Logarithmic Mappings

Logarithms simplify multi-variable algebraic expressions using three main operational rules:

$$\log_b(M \cdot N) = \log_b(M) + \log_b(N) \quad (\text{Product Property}) \quad (5.3)$$

$$\log_b\left(\frac{M}{N}\right) = \log_b(M) - \log_b(N) \quad (\text{Quotient Property}) \quad (5.4)$$

$$\log_b(M^p) = p \cdot \log_b(M) \quad (\text{Power Property}) \quad (5.5)$$

6 | Systems of Equations and Matrix Transformations

When multiple variable conditions must be satisfied simultaneously, they are organized into structured systems of linear equations.

6.1 Linear Systems Analysis

A system of equations identifies where multiple spatial lines intersect. These intersection points can be calculated using substitution, row elimination, or by transforming the system into an augmented matrix for systematic Gauss-Jordan row reduction.

6.2 Matrix Algebra and Inversion

We can bundle linear systems into a concise matrix equation:

$$A\mathbf{x} = \mathbf{b} \tag{6.1}$$

Where A is the coefficient matrix, $\mathbf{x}$ is the target variable vector, and $\mathbf{b}$ is the output constant vector. If the matrix determinant is non-zero ($|A| \neq 0$), the matrix is invertible, allowing us to compute the unique solution vector directly:

$$\mathbf{x} = A^{-1}\mathbf{b} \tag{6.2}$$

7 | Analytic Geometry

Analytic geometry uses coordinate equations to define geometric shapes and pathways.

7.1 Conic Sections Canonical Frameworks

Conic sections map out the two-variable quadratic equations produced when a plane slices through a three-dimensional double cone.

- **Circle:** Points at a constant radius r from a center (h, k) :

$$(x - h)^2 + (y - k)^2 = r^2 \quad (7.1)$$

- **Ellipse:** An elongated circular orbit with major and minor scaling axes a and b :

$$\frac{(x - h)^2}{a^2} + \frac{(y - k)^2}{b^2} = 1 \quad (7.2)$$

- **Hyperbola:** A split open-curve structure tracking two symmetric paths:

$$\frac{(x - h)^2}{a^2} - \frac{(y - k)^2}{b^2} = 1 \quad (7.3)$$

8 | Right Triangle Trigonometry

Trigonometry establishes fixed structural ratios linking internal triangle angles to side dimensions.

8.1 Right Triangle Trigonometric Projections

Given a right-angled triangle anchored by an acute reference angle θ :

$$\sin \theta = \frac{\text{Opposite}}{\text{Hypotenuse}}, \quad \cos \theta = \frac{\text{Adjacent}}{\text{Hypotenuse}}, \quad \tan \theta = \frac{\text{Opposite}}{\text{Adjacent}} \quad (8.1)$$

The reciprocal trigonometric operators are defined as:

$$\csc \theta = \frac{1}{\sin \theta}, \quad \sec \theta = \frac{1}{\cos \theta}, \quad \cot \theta = \frac{1}{\tan \theta} \quad (8.2)$$

9 | Periodic Functions and the Unit Circle Engine

To evaluate angles beyond a simple 90° right triangle, trigonometry is mapped onto a continuous coordinate tracking loop.

9.1 The Unit Circle Space

The unit circle centers on the coordinate origin with a fixed radius of 1: $x^2 + y^2 = 1$. Rotating a unit vector counterclockwise by an angle θ maps the coordinate points perfectly to trigonometric values:

$$(x, y) = (\cos \theta, \sin \theta) \quad (9.1)$$

9.2 Continuous Wave Oscillation

Because a circle repeats its path every 2π radians (360°), sine and cosine functions create continuous, indefinitely oscillating wave outputs:

$$\sin(\theta + 2\pi k) = \sin \theta, \quad \cos(\theta + 2\pi k) = \cos \theta \quad (k \in \mathbb{Z}) \quad (9.2)$$

10 | Trigonometric Identities and Verification

Trigonometric identities establish algebraic rules for converting, simplifying, and verifying complex wave equations.

10.1 Fundamental Pythagorean Identities

Derived directly from the unit circle equation ($x^2 + y^2 = 1$), these structural anchors provide immediate algebraic conversions:

$$\sin^2 \theta + \cos^2 \theta = 1, \quad 1 + \tan^2 \theta = \sec^2 \theta, \quad 1 + \cot^2 \theta = \csc^2 \theta \quad (10.1)$$

10.2 Compound Expansion and Double-Angle Adjustments

When dealing with combined or doubled angle parameters, use these exact expansion properties:

$$\sin(\alpha \pm \beta) = \sin \alpha \cos \beta \pm \cos \alpha \sin \beta \quad (10.2)$$

$$\cos(2\theta) = \cos^2 \theta - \sin^2 \theta = 2 \cos^2 \theta - 1 = 1 - 2 \sin^2 \theta \quad (10.3)$$

$$\sin(2\theta) = 2 \sin \theta \cos \theta \quad (10.4)$$

11 | Advanced Trigonometric Applications

Trigonometric systems extend beyond right triangles to map oblique spaces, polar tracking systems, and spatial vectors.

11.1 Oblique Triangle Mapping

When solving triangles that lack a convenient 90° corner, use these general geometric laws:

- **Law of Sines:** Useful when matching known sides to their opposite angles:

$$\frac{a}{\sin A} = \frac{b}{\sin B} = \frac{c}{\sin C} \quad (11.1)$$

- **Law of Cosines:** Extends the Pythagorean theorem to accommodate any internal corner angle:

$$c^2 = a^2 + b^2 - 2ab \cos C \quad (11.2)$$

11.2 Polar Coordinates and Spatial Vector Grids

Polar systems track an object's position using a straight-line radial distance and a rotational angle (r, θ) , shifting away from traditional rectangular grids (x, y) :

$$x = r \cos \theta, \quad y = r \sin \theta, \quad r = \sqrt{x^2 + y^2} \quad (11.3)$$

Spatial vectors use this polar framework to track multi-dimensional forces, target velocities, and navigation paths.

12 | Sequences, Quantified Probability, and Combinatorics

Discrete mathematics models ordered lists, counting arrangements, and quantified likelihood indices.

12.1 Discrete Sequences and Infinite Series

Ordered chains of data follow specific structural patterns:

- **Arithmetic Sequences:** Each term advances by adding a constant step difference d :

$$a_n = a_1 + (n - 1)d \quad (12.1)$$

- **Geometric Sequences:** Each term scales by multiplying a constant ratio r :

$$a_n = a_1 r^{n-1} \quad (12.2)$$

12.2 Combinatorics Logic and Arrangements

When selecting a subset of items from a total pool n , the counting logic depends on whether order matters:

- **Permutations:** Use this formula if the exact sequence order is critical:

$$P(n, r) = \frac{n!}{(n - r)!} \quad (12.3)$$

- **Combinations:** Use this formula if the subset grouping is order-independent:

$$C(n, r) = \frac{n!}{r!(n - r)!} \quad (12.4)$$

12.3 Axiomatic Probability

The probability score of a target event A occurring within a uniform sample space S is quantified as the ratio of favorable outcomes to total possibilities:

$$P(A) = \frac{n(A)}{n(S)} \quad \text{clamped strictly to the domain: } 0 \leq P(A) \leq 1 \quad (12.5)$$

13 | Gateway to Calculus: Continuous Limits

Calculus bridges the gap between static algebra formulas and dynamic, continuously changing processes.

13.1 The Concept of a Limit

A limit tracks the output trend of a function as its input variable draws closer to a target value c , without needing to evaluate the function exactly at that point:

$$\lim_{x \rightarrow c} f(x) = L \quad (13.1)$$

13.2 Mathematical Continuity

A function $f(x)$ is continuous at a point $x = c$ if its graph flows smoothly through that point without gaps, jumps, or vertical breaks. This requires the limit trend to match the exact value of the function:

$$\lim_{x \rightarrow c} f(x) = f(c) \quad (13.2)$$

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